Supplementary Material

The works cited here are listed under References at the end of this file. File paths refer to the GitHub repository named in the Data Availability Statement of the main text, and ledger IDs are rows of `docs/paper-prep/results_ledger.csv` in that repository.

S1. Models checked

We ran the model in float16 with float32 output projections, using Python 3.12, PyTorch 2.8.0 with CUDA 12.8 and Transformers 4.57.6 on NVIDIA RTX PRO 6000 graphics processors with 96 GB of memory, and computed statistics with NumPy 2.5.3, pandas 3.0.6 and SciPy 1.18.1. Because routing and floating-point arithmetic make a score depend on the other items in its batch, every score was averaged over three batch layouts unless a figure notes one.

The inclusion criterion of Section 2.1 was applied in two forms. The original form, frozen on 2026-09-27, first required string-matching accuracy of at least .80 and order consistency of at least .90 on the string items. It then required point accuracy of at least .70 on the high-demand items with a bootstrap lower limit above .50. It also required semantic order consistency of at least .90 and a control effect with a lower limit above 0.20 logits. The revised form, frozen on 2026-09-28 after the first candidate models had been screened, scores an item as correct when its margin averaged over the two option orders is positive. It keeps the .50 lower limit and the 0.20 logit control effect and drops the .70 point threshold and the consistency gates. Order consistency is reported but no longer decides.

The control effect E is the predicted margin change across the fixed range of TDS. It comes from the regression of the order-averaged margin on TDS with task as a covariate, $E = 0.615\ \beta_{\mathrm{TDS}}$, fitted on all 160 feature-matching items with intervals from resampling concept groups. The high-demand items are the 80 color and size items whose TDS lies below the within-task median, which in these norms coincides with the 80 items of negative TDS (40 per task).

The four candidates examined in the October 2026 census first passed a format check. Each saw 24 demonstration problems in both option orders, 48 presentations, and had to produce the requested label as its first answer on at least .90 of them. The thinking model had a 2,048-token budget and was judged on the answer after

its closing thinking marker. The gpt-oss model failed its first format check. A revised format added an "Answer:" prefix in the Harmony assistant channel, and its criterion results belong to that format.

Table S1 lists the thirteen models. Ten failed, two passed and one was checked for format only. Every failing model failed on high-demand accuracy, and every one of them had a control effect above 0.20 logits at its lower limit. The earlier Qwen3-30B-A3B release passed the revised criterion within the census branch, and no component search followed in it.

Table S1. Models checked against the inclusion criterion. Accuracy is the proportion of the 80 high-demand items answered correctly, with 95% bootstrap intervals over concept groups. E is the control effect in logits over the 160 feature-matching items. Rows marked † report accuracy under the original scoring (each presentation scored on its own), because no order-averaged value was recorded for those runs. String is accuracy on the 160 string-matching items. Revisions are the first seven characters of the pinned commit. Source rows are IDs of `docs/paper-prep/results_ledger.csv` with the prefix omitted (`g1_olmoe-`, `g1_qwen-`, `g1_phi-`, `g1_deepseek-`, `g1_gptoss_p12b-` or `title_gap_model_census-`), and models without ledger rows name their result file.

Model (revision)	Rule applied	High-demand accuracy [95% CI]	E, logits [95% CI]	String	Outcome	Source rows
OLMoE-1B-7B-0924 base (6d84c48)	original	.156 [.101, .215] †	1.31 [1.11, 1.52]	.48	failed	`22c6fc9c6d13`, `d2eb35bdd2a3`, `3b413e682bf1`
OLMoE-1B-7B-0924 instruct (7f1c97f)	original	.25 [.19, .31] †	4.60 [3.80, 5.48]	.55	selected within the pair, failed	`302876345f9e`, `ea471814e3ec`, `e1225f130df0`
OLMoE-1B-7B-0125 base (9b0c1aa)	original, backup	.431 [.388, .468] †	0.71 [0.62, 0.81]	.57	selected within the pair, failed	no ledger row, `reports/g1_backup/g1_result.json`
OLMoE-1B-7B-0125 instruct (b89a7c4)	original, backup	.319 [.263, .373] †	0.48 [0.38, 0.60]	.52	failed	no ledger row, `reports/g1_backup/g1_result.json`

Model (revision)	Rule applied	High-demand accuracy [95% CI]	E, logits [95% CI]	String	Outcome	Source rows
OLMo-2-13B instruct (3a5c85b)	original, exploratory	.356 [.293, .420] †	1.14 [0.91, 1.39]	.64	failed	no ledger row, `reports/e0_explore/olmo2-13b-instruct/g1_result.json`
Qwen3-30B-A3B-Instruct-2507 (0d7cf23)	revised	.625 [.519, .734]	5.54 [3.79, 7.21]	.88	passed, main model	`7d88b53dc3fa`, `3098831290e2`, `541f4b1894d0`
Phi-3.5-MoE instruct (4368845)	revised	.388 [.288, .494]	1.57 [1.13, 2.02]	.82	failed	`1b7e6f5a3ca6`, `5afd74c5b999`, `6aac651c2678`
DeepSeek-V2-Lite chat (8586474)	revised	.238 [.150, .333]	0.33 [0.22, 0.46]	.50	failed	`8c85c24781f0`, `3039aa1e130f`, `ab54d777f260`
gpt-oss-20B, revised format (6cee5e8)	revised	.263 [.175, .359]	1.57 [1.28, 1.87]	.64	format passed, failed	`01939fd9c049`, `413fd17d8258`, `44723502d67d`
Qwen3-Coder-30B-A3B-Instruct (b2cff64)	revised, census	.475 [.367, .588]	7.10 [5.49, 8.68]	.83	format 48 of 48, failed, replication stopped	`3bd18baa72b9`, `107a300f5bb1`, `da2a4e75543f`, `5b5de0eb3d01`
Qwen3-32B, nonthinking (9216db5)	revised, census	.388 [.280, .500]	8.13 [7.18, 9.08]	.92	format 48 of 48, failed	`3ca4a6a04871`, `c0a679cb6612`, `6b3cbd5b38a3`, `675e80b7065f`
Qwen3-30B-A3B-Thinking-2507 (144afc2)	format only	not run	not run	not run	format 44 of 48 passed, 3 budget exhausted	`31bf4591e2b3`, `a707769b3d35`, `f41ad03c9f61`

Model (revision)	Rule applied	High-demand accuracy [95% CI]	E, logits [95% CI]	String	Outcome	Source rows
Qwen3-30B-A3B, earlier release, nonthinking (ad44e77)	revised, census	.663 [.556, .769]	4.23 [2.79, 5.75]	.88	format 48 of 48, passed, census only	`187974614aee`, `fcf09e7cb9a0`, `072c91bbfb1d`, `8082bdaa2669`

Synonym judgment accuracy, where recorded, was .93 for the main model, .57 and .65 for the two OLMoE-0924 models, .93 for Phi-3.5-MoE, .51 for DeepSeek-V2-Lite and .85 for gpt-oss-20B. The main model's order consistency was .74 on the feature-matching items and .86 on the string items. Model identities and pinned revisions come from `src/lesion/models.py` and `src/lesion/model_census.py`. The main model was loaded in float16 with float32 output projections and scoring, under Python 3.12, PyTorch 2.8.0 with CUDA 12.8 and Transformers 4.57.6.

S1.1. Exploratory second model: Phi-3.5-MoE

Phi-3.5-MoE instruct showed a control effect of 1.57 [1.13, 2.02] logits but scored .39 [.29, .49] on the high-demand items, below chance, so it failed the inclusion criterion (Table S1). We nevertheless ran the component search and the dissociation test on it, as an exploratory check on whether the division is particular to the main model. Phi ran in float16, because the check that shared-prefix scoring reproduces full scoring did not fully pass in either precision and float16 had the smaller noise floor. The control search selected four heads, 0:2, 1:1, 6:11 and 6:28. On the held-out items their lesion enlarged the control effect by 6.22 [4.78, 7.62] logits, more than any of 200 matched random sets (median 0.31, maximum 1.85), changed the frequency effect by -0.63 [-2.47, 1.28] nats and improved the string-matching items by 2.26 [1.36, 3.11] logits. Unlike SC4, it also improved naming slightly, by 0.44 [0.31, 0.57] nats. The retrieval search selected four experts, 3:7, 5:10, 26:6 and 27:8, and four heads, 10:5, 14:21, 17:9 and 18:26. Lesioning all eight cost naming 21.43 [20.59, 22.31] nats and raised the loss on neutral text by 163%, and the generated answers broke down. Lesioning expert 3:7 or 5:10 alone raised the neutral-text loss by 31% or 37%, because these two experts act at the first token of every passage, and the model cannot lose them. Without them, the remaining six components impaired rare targets 0.78 [0.35, 1.23] nats more than common ones on the held-out concepts, more than any of 100 matched random sets (maximum 0.29), cost naming 0.21 [0.00, 0.44] nats overall and raised the neutral-text loss by 2.0%. The

frequency contrast between the six components and the four control heads was 0.67 [0.18, 1.16] nats. The model's intact accuracy on the hard items limits what its lesions can show.

We then gave the two Phi sets the naming battery of Results 3.2 and 3.3, with the same 160 held-out concepts, prompts, scorers, batch layouts and mean ablation, and with Phi's output-layer bias added to every score (`experiments/second_model/run_phi_battery.py`, plan and reading rules fixed before the run in `docs/plans/phi-battery.md`). PR6 denotes the six retrieval components left after the two experts Phi cannot lose, and PC4 the four control heads. The intact model chose the target in the six-option choice and preferred it over the associate in naming on 156 of the 160 concepts, and the results below are for these. Lesioning PR6 lowered naming by 0.17 [0.03, 0.32] nats and left word-to-definition matching intact, improving it by 0.93 [0.77, 1.09] logits, and naming fell 0.18 [0.06, 0.31] nats more after the complete PR6 lesion than after the lesion that kept 75% of its output. The joint lesion added in naming, with an interaction of -0.00 [-0.06, 0.06] nats. These three results match the main model. Lesioning PC4, by contrast, improved the six-option choice by 0.44 [0.23, 0.65] logits instead of impairing it, so the double dissociation of naming and six-option choice did not hold. Restricted to the answer positions, the PR6 lesion left naming unchanged at 0.00 [-0.08, 0.08] nats, and the PC4 lesion changed the miscue cost minus the cue benefit by -0.10 [-0.23, 0.03] nats, against 0.44 for SC4 in the main model. On the 100 cueing concepts the PC4 value, -0.17 [-0.34, -0.01], lay within the range of 20 matched random sets of four heads (maximum 0.18). Lesioned at every position, PC4 raised the benefit of a correct cue by 1.84 [1.49, 2.21] nats, whereas PR6 changed it by -0.13 [-0.52, 0.26], so in Phi the control lesion, not the retrieval lesion, raised the cue benefit, the pattern of SA that the main model did not show (H4a, Results 3.4). In the generated answers to the 240 held-out naming concepts, PR6 lowered accuracy from .57 to .45 for rare names and from .83 to .77 for common ones, a difference of .06 [-.03, .14]. The neutral-text loss rose by 2.0% under PR6 and 0.8% under PC4. In sum, the Phi retrieval set reproduced the main model's retrieval profile in naming and matching, whereas its control heads did not reproduce the main model's control profile on the battery.

Phi's output layer has a bias term (32,064 values, standard deviation 0.019 logits), which the shared scorers of the earlier Phi runs left out. Scored with and without it, the frequency effect of PR6 on the held-out naming concepts was 0.787 and 0.785 nats, so the omission does not change the results reported above.

S2. Materials

S2.1. Semantic control judgments

The 480 items of Wu and Hoffman (2022) comprise 80 color, 80 size, 160 synonym and 160 string-matching judgments, forming 472 concept groups because some color and size items share a probe. Similarities, TDS and human accuracy fields come from the original materials, and word frequency from the supplied SUBTLEX-UK norms. The split by concept group used seed 0 and stratified control demand or frequency. Screening holds 190 groups and 194 items, development 93 groups and 94 items, and the held-out test 189 groups and 192 items. The confirmatory control test used the 63 color and size test items, 30 of them of high control demand at the fixed within-task median of TDS. The regression of damage on distractor similarity used the standardized word-vector similarity of distractor and probe (field w2vD1P) from `data/materials/m1/associations.csv`. Each item was presented in both option orders, and label scores were averaged over batch sizes 400, 128 and 48, in which the shared prefix of a task is computed once.

S2.2. Naming from definition

The 600 concepts were assembled in two batches of 300 with seeds 0 and 1. Each batch sampled six SUBTLEX-US Zipf bins with boundaries 1, 2, 2.5, 3, 3.5, 4 and 7.5. A concept was eligible when its primary sense was a concrete nonhuman noun and its target was one word. Its WordNet gloss could not contain the target's stem, and its single-word synonyms had to be sufficiently rarer than the target. Four demonstration concepts were kept apart. The split holds 240 screening, 120 development and 240 test concepts, with 40 test concepts in each bin. WR selection used the 120 screening concepts of the first batch, at commit d755a4e, and the rare versus common boundary stayed at the Zipf median of that batch, 2.99965. The intact model's scores on all 600 concepts were inspected for validity before the freeze, and no concept was removed. The definitions are WordNet glosses, so no item of this set was drafted by an AI tool. The remaining word pool could not supply a second held-out set of the same design, so this test was run once.

S2.3. Naming battery

The battery holds 160 test concepts, two calibration sets of 40 concepts (a and b) and 10 reserve concepts, and its final file retains one excluded row. Each definition was written from its WordNet entry, and the coordinate and superordinate were grounded in WordNet categories. Associates came from the Small World of Words English first-response norms under the frozen association rule, and unrelated words were matched in frequency and length. American spellings were used. Targets of the earlier screening materials and demonstrations were excluded, and set b

balanced whether the target or the associate was the more frequent word. The keep and drop decisions of the review (2026-10-07) are in `data/materials/m3/curation.csv`, with sense checks against the control items in `data/materials/m3/m1_sense_review.csv`. The drafting log is `data/materials/m3/items_log.json` and the final allocation `data/materials/m3/final_log.json`.

Three validity rules were fixed on the intact model before the freeze. For error types, the target had to be the most probable of the five candidates and hold at least half of their probability mass, which left 156 of the 160 concepts. The four excluded were m000 percolator, m043 shortcake, m081 waffles and m124 donuts. For cueing, free naming required a positive target versus associate log-probability difference, similar token lengths and different first two letters, which left the 100 concepts marked free_naming. The progressive cueing comparison used the broader free_naming_ok field, 132 concepts. Word-to-definition matching used one foil per concept, 156 from the same category and 4 from the same domain, generated by rule before the freeze. The general weakening model was fitted on the 40 concepts of calibration set b. The mechanism explorations that produced the secondary hypotheses used the 80 calibration concepts. Candidate scores were averaged over batch sizes 16, 11 and 7, and label scores over 32, 21 and 13.

S2.4. Semantic families

Each family gives one concept naming from definition, synonym choice, choice between the correct definition and a close alternative, copying and word production from options. A true-or-false property judgment was added where an independent property could be specified. Senses and accepted synonyms are single-word WordNet 3.0 lemmas of the recorded synset, frozen before any model answer. Frequency is the lg_BNC_freq field and semantic diversity the SemD field of the Hoffman et al. (2013) norms, matched by exact normalized word. The original set holds 78 screening, 43 development and 63 test families, split with seed 1910 so that connected stems and identical synsets stay in one split, stratified by frequency and semantic diversity. An extension reviewed separately added 57 test families with seed 1921, chosen in fixed seeded order within frequency bins and excluding every stem of the earlier materials and of the homographs. The final set holds 120 test families, 40 per frequency band, 44 with a property judgment. The accepted material totals 241 families, and every rejected draft is retained in the material records (184 active and 176 excluded rows in the original pool).

The families were produced by the pipeline `experiments/title_gap/prepare_fine.py` and reviewed in two rounds of source checking and semantic review before any model score on the new material. The extension review trace is `.aris/traces/experiment-audit/2026-10-19-fine-extension/response-full.md` (SHA-256 e6054f4a…), with the review records

`review-record.json` and `review-record-final.json` beside it, four first-round and one second-round replacements recorded in the manifest. The review records of the original pool are `REPORT.md`, `fine-exclusions.csv` (the first round), `fine-revisions.csv`, `fine-revisions-final.csv` and `review-trace.json` in `reports/explore/2026-10-19-title-gap-material-audit/`.

S2.5. Homograph families

Noonan et al. (2010) found that patients with SA understood homonyms better when a sentence cued the relevant meaning and worse when it cued the other meaning, and Hoffman et al. (2018) simulated this test in their model. We adapted the test with 80 homographs that have two normed meanings, 40 for development and 40 for the final test. The model saw each homograph with a weak or a strong description of one meaning. It then produced an associate of that meaning with no cue, a correct cue or a misleading cue, and in a choice version it chose between a word for each meaning (Section S2.7). A separate search on the development families selected eight production control components (PC8), whose lesion weakened the model's preference for the context-supported answer (Section S3.8). Generated answers were scored against the target, its WordNet synonyms and human association responses, and answers that no rule or judge could classify enter the results as a range, counted first as wrong and then as right (classification bounds).

The 80 homographs and their two normed meanings come from the spoken British English norms of Gilbert and Rodd (2022), DOI 10.5334/joc.194, files uk_norms.csv and uk_responses.csv. The split into 40 development and 40 final families used seed 1910 and kept connected Porter stems of the ambiguous words and candidate answers together. Each meaning has low and high contextual support and no cue, correct cue and misleading cue conditions. In place of the patients' sentence, the model saw the word and a description of the intended meaning and produced one word associated with that meaning. For the lightning meaning of bolt, the weak description was "in stormy weather" and the strong one "a sudden jagged flash of electrical discharge from the sky". The correct cue gave the first two letters of an associate of the intended meaning, and the misleading cue gave those of an associate of the other meaning. In the choice version of the same trials, the word, the description and the cue line were followed by two candidate words labeled A and B, one associated with each meaning, the model answered with the letter, and each pair was presented in both orders. Low and high support differ in semantic specification, a broad domain against a detailed meaning, with equal word counts through neutral padding. The production control search later divided the development families into 20 screening and 20 validation families with seed 1920. Generated answers were classified against the human association responses, and the response lexicon is not redistributed.

S2.6. Neutral text and licenses

Reference means came from 200 WikiText-103 validation passages chosen with seed 0 and truncated to 512 tokens. Neutral-text loss for the general damage checks used 40 passages, the first 256 tokens of each in the graded battery and the first 128 tokens of each in the family runs. WordNet is redistributed with its notices. SWOW, the semantic diversity workbook and the homograph response files are supplied by reference.

S2.7. Prompts and presentation

Every prompt of the five item sets has the same shape: an instruction, then the demonstration items where the set has them, then the test block, joined by blank lines. For the main model the reply rule quoted below was appended to the instruction, and the whole text was placed in one user turn of the model's chat template with no system message. The template writes the prompt as `<|im_start|>user\n{prompt}<|im_end|>\n<|im_start|>assistant\n`, so the first answer token is the first token of the assistant turn. For the base models of the pilot the text was preceded by `<|endoftext|>`, no reply rule was added, and the answer followed the final "Answer:" with a leading space. Label scores are the float32 logits of the label tokens at the last prompt position, the position whose hidden state predicts the first answer token. The labels are the bare tokens `A` and `B` (and `C` to `F` in the six-option task) for the main model and the same letters preceded by a space for base models. Candidate scores place the candidate word as the content of the assistant turn, feed prompt and answer together (teacher forcing) and sum the log-probabilities of the answer tokens, each read at the position that predicts it, with the output projection in float32. Generated answers come from greedy decoding of the same prompt (`do_sample=False`), left padded in batches of 16, with the token limits given below, and special tokens are stripped from the decoded text.

Semantic control judgments. The template file gives one instruction and one block per task. The main instructions were used throughout, and the alternatives alt1 and alt2 of the file were never used.

```
instructions:
  colour:
    main: "Choose the word that is closer to the first word in colour."
  size:
    main: "Choose the word that is closer to the first word in size."
  synonym:
    main: "Decide whether the two words have similar meanings."
  string:
    main: "Choose the string that is closer to the first string in
    ↪  spelling."
```

```
blocks:
  colour: "Word: {probe}\nA: {a}\nB: {b}\nAnswer:"
  size: "Word: {probe}\nA: {a}\nB: {b}\nAnswer:"
  string: "String: {probe}\nA: {a}\nB: {b}\nAnswer:"
  synonym: "Words: {word1}, {word2}\nA: {a}\nB: {b}\nAnswer:"
synonym_options:
  similar: "similar in meaning"
  different: "different in meaning"
demo_answer: " {label}"
instruct:
  reply_rule: " Reply with a single letter: A or B."
```

The probe line is the cue of this set: the probe word, the probe string, or the two words of a synonym item. In the synonym task the two options are the fixed phrases "similar in meaning" and "different in meaning", so its A and B name the response options themselves. Each task has six demonstration items, 24 in all, in `data/materials/m1/demos.csv`, shown in their recorded order with the answer letter after "Answer:". Each test item was presented twice, with the first option at A and then the second option at A. The demonstrations stayed fixed across the two orders, except in the synonym task, where the demonstration options and their answer letters swapped with the test item (A, B, B, A, B, A became B, A, A, B, A, B). The score of a presentation is the logit of the correct label minus the logit of the incorrect label at the first assistant token, and the margin of an item is the mean over its two orders. No generated answers were scored in this set. The format check of the census models (Section S1) generated at most 4 new tokens greedily, and 2,048 for the thinking model.

Naming from definition. The instruction, the reply rule and the block are:

```
INSTRUCTION = "Name the single English word that each definition
↪  describes."
REPLY_RULE = " Reply with the word only, in lowercase."
BLOCK = "Definition: {definition}\nWord:"
```

A demonstration is the block followed by a space and the target word. The four demonstration concepts, engine, hotel, kitchen and meat, with their WordNet glosses, are in `data/materials/m3_knowledge/demos.csv` and precede every test definition in that order. There are no options and no cue. The scored answer is the assistant turn as the chat template writes it, the word's tokens followed by the end-of-turn token `<|im_end|>`, and its summed log-probability is the naming score. The number of word tokens and the rank of the word's first token among the whole vocabulary at the answer position were recorded with it. Scores were averaged over batch sizes 16, 11 and 7 in the confirmatory test, and the random sets were scored at batch size 16. No answers were generated in this set.

Naming battery. Free naming, cued naming and progressive cueing use:

```
NAMING_INSTRUCTION = "Name the thing that matches each description."
NAMING_REPLY_RULE = " Reply with the word only, in lowercase."
HINT = 'Hint: the word starts with "{xx}".'
block: "Description: {definition}" + ["\n" + HINT] + "\nAnswer:"
```

The hint line, when present, stands between the description and "Answer:". The cue gives the target's first two letters and the miscue the associate's first two letters. Three demonstration concepts are fixed in the code and occur nowhere else in the materials:

```
clock: "an object on a wall or desk with a numbered face and moving hands
↪  that show the time"
kite: "a light frame covered with paper or plastic that children fly high
↪  on a long cord"
elevator: "a moving compartment that carries people up and down between
↪  the floors of a building"
```

In a hint condition the demonstrations carry correct hints of the same kind, and without a hint they carry none. In the progressive cueing test the hint gives the target's first j letters, j from 0 to one less than the word's length, and the demonstrations' hints have the same length, capped at all but the last letter of the demonstration target. The candidates scored are the target and the associate for cueing and all five candidates for error types. The scored tokens are the word's tokens alone, which start the assistant turn without a leading space, and the end-of-turn token is not scored. The six-option choice uses:

```
CHOICE_INSTRUCTION = 'Choose the word that the description defines. If
↪  none of the words fits, choose "{ct}".'
CHOICE_REPLY_RULE = " Reply with the letter only."
CANNOT_TELL = "cannot tell"
LETTERS = ("A", "B", "C", "D", "E", "F")
block: "Description: {definition}\nA. {w}\nB. {w}\nC. {w}\nD. {w}\nE.
↪  {w}\nF. {w}\nAnswer:"
```

The six options are the target, coordinate, superordinate, associate, unrelated word and "cannot tell". Their base order for an item is a permutation drawn from a random generator seeded with the SHA-256 of the item id, and each item was presented in the six cyclic rotations of that order, so every option sat once at every letter. The three demonstrations show their options in a fixed order, with the answer letter after "Answer:", and "cannot tell" is never the demonstration answer:

```
clock:    ("watch", "clock", "pillow", "timepiece", "cannot tell",
↪  "alarm")         answer B
kite:     ("plaything", "sky", "doll", "cannot tell", "kite", "onion")
↪  answer E
elevator: ("stairs", "cannot tell", "machine", "elevator", "blanket",
↪  "escalator")  answer D
```

The six label logits were read at the first assistant token and averaged over the label batch sizes of Section S2.3. The probability of an option is the softmax over the six labels within a rotation, averaged over the rotations. The other formats of Section 2.2.3 use the same demonstrations:

```
options_word:       CHOICE_INSTRUCTION + " Reply with the chosen word
↪  only, in lowercase."
                    the six-option block in the base order only,
                    ↪  demonstrations answered with the target word
word_to_definition: "Choose the description that defines the given word."
↪  + " Reply with the letter only."
                    "Word: {target}\nA. {definition}\nB.
                    ↪  {definition}\nAnswer:"
copy:               "Repeat the given word exactly." + " Reply with the
↪  word only, in lowercase."
                    "Word: {target}\nAnswer:"    demonstrations "Word:
                    ↪  clock\nAnswer: clock" and so on
```

In word-to-definition matching the demonstrations pair clock with the kite definition (answer A), kite with the elevator definition (answer B) and elevator with the clock definition (answer A), and each test concept was presented with its own definition at A and then at B. The hinted-demonstration baseline of the mechanism analyses uses the naming instruction with correctly hinted demonstrations and no hint on the test item. Generated answers of the battery were produced for free naming and for every cue length, greedy with at most 6 new tokens in the battery and cueing runs on the calibration and test concepts, and at most 8 new tokens in the clinical cueing runs of Section S3.11.

Semantic families. These prompts carry no demonstrations, and each contains its own instruction and reply rule:

```
naming:      "Name the thing or concept described. Reply with one word
↪  only.\nDescription: {definition}\nAnswer:"
copying:     "Repeat this word exactly. Reply with the word only.\nWord:
↪  {target}\nAnswer:"
definition:  "Which description defines {target}? Reply with the letter
↪  only.\nA. {a}\nB. {b}\nAnswer:"
synonym:     "Which word can mean the same as {target}? Reply with the
↪  letter only.\nA. {a}\nB. {b}\nAnswer:"
```

```
property:    "For {target}, is this claim true? {claim}\nReply with the
↪  letter only.\nA. {a}\nB. {b}\nAnswer:"
word answer: "Which word can mean the same as {target}?\nChoose between
↪  these two words: {a}, {b}.\nReply by repeating the chosen word only.
↪  Do not add a letter or explanation.\nChosen word:"
```

Naming scores the target, its synonym and the close foil (the coordinate) as candidates, and copying scores the target. Definition choice sets the target's definition against the foil's definition, and synonym choice sets the synonym against the foil, each in both orders, the correct option at A and then at B. The property item presents the true claim and the false claim, each with the options "true" and "false" in both orders, four prompts per family. The word-answer format names the synonym and the foil in the question in both orders and scores the two words as candidates. The cue of this set is the target word itself in the comprehension prompts and the definition in naming. Label logits were read at the first assistant token and averaged over batch sizes 16, 11 and 7, and the margin was taken after averaging. Candidate scores used batch sizes 16, 11 and 7 in development and 64, 47 and 31 in the final run. Every candidate-scored prompt also received a greedy answer of at most 8 new tokens.

Homograph families. These prompts carry no demonstrations. Production and choice share one body:

```
body:       "Word: {word}\nIntended meaning: {context}" + [hint]
hint:       '\nAn additional hint suggests a word starting with "{xx}".
↪  The hint may be wrong.'
production: "Give one word closely associated with the intended meaning.
↪  Reply with that word only.\n" + body + "\nAnswer:"
choice:     "Choose the word associated with the intended meaning. Reply
↪  with the letter only.\n" + body + "\nA. {a}\nB. {b}\nAnswer:"
```

The context is the low or high support description of the intended meaning, and the hint line is present only in the cue conditions. The cue "{xx}" is the first two letters of the intended meaning's associate (correct) or of the other meaning's associate (misleading). Production scores the two associates as candidates, the intended meaning's as target and the other meaning's as competitor, and generates a greedy answer of at most 8 new tokens. The choice version lists the same two words at A and B, in both orders, and reads the two label logits at the first assistant token. Each meaning therefore has six production prompts and twelve choice prompts, from two support levels, three cue conditions and two orders. Batch sizes were 16, 11 and 7 for both kinds of score.

S3. Lesions

S3.1. Locating the components

Attribution patching ranked all 6,144 experts and 1,536 heads, and the 32 highest-ranked candidates in each direction were ablated one at a time in three batch layouts. Effects were standardized within the verified shortlist, and each component was assigned to the direction with the larger standardized effect, ties to control. Control candidates were then ranked by raw control effect, and a candidate whose string-matching damage exceeded its control effect was removed as domain general. Retrieval candidates were ranked by the extra naming damage for rare targets, with token count as a covariate. The initial four were expanded once to the eight with the largest verified effect, the expansion that the study plan had provided for, and WR8 was fixed before its test. The string exclusion was not applied to the retrieval search.

Matched random sets preserve the number, types and layers of a target set. A random expert must lie within .25 of the target expert's within-layer routing-frequency percentile. Members of WR8 and SC4 cannot be sampled, and each set is lesioned as a whole. The naming random sets were scored in the single batch layout of 16, in which the WR8 effect was 1.12 nats. The random tail probability is $(k + 1)/(N + 1)$. Development checks used 20 sets per side, and the graded battery reused three sets per side for descriptive random single and mixed lesions, listed in `docs/analysis-plan/optogenetics.md` section 2.

S3.2. Mean ablation

The lesion code was checked with automated tests on small random models.

Expert ablation acts on the expert's output before multiplication by its routing weight, and head ablation on the head's slice of the input to the attention output projection. Expert means average the activations of the tokens routed to that expert, head means average all valid positions, accumulated in float32 with padding excluded. The means file has SHA-256 98b3865a71b0c8e85c530cb78d5438f9223a828fdf77c907315785b1f1de0d52. On the neutral text 1,839 experts received fewer than 500 tokens, so their means are coarser. Graded ablation retained .25, .5 and .75 of the output, and amplification used $\alpha = 1.5$ and 2. Zero ablation, $h' = \alpha h$ with $\alpha = 0$, was run on the development items. It gave a frequency contrast of 0.76 [0.10, 1.48] nats on 120 naming concepts and a control contrast of 8.28 [5.49, 11.15] logits on 31 color and size items, both exploratory.

S3.3. Position-restricted lesions

A Boolean mask over the current token positions divides each prompt into four regions. The context region holds the instructions and demonstrations. The probe

region holds the target's description or the word itself. The middle region holds the intervening cue, options and template text. The answer region begins at the last prompt position, whose hidden state predicts the first answer token, and includes the answer tokens used in candidate scoring. Positions outside the chosen regions keep their intact values. The answer-region recovery (H2e) ablated WR8 at the answer region and then retained .25, .5 and .75 of its output there, with all other positions intact, at commit 750d8a4.

The confirmatory graded and positional battery comprised 45 states. Four were intact, WR8, SC4 and the joint lesion. Fourteen were WR8 and SC4 at α = .25, .5, .75, 1.5 and 2, and one random set per side at α = .5 and 2. The rest were three recovery states, twelve regional states and twelve random single and mixed lesions. The regional states were SC4 at the probe, middle, answer, probe and middle, probe and answer, middle and answer, all three, and the context region, and WR8 at each single region. A rule was not interpreted if a state in it raised neutral-text loss by more than 10%. Regional states cannot be checked on plain text, so the whole-sequence state of the same set served as a proxy bound. The relative rise was 1.3% for WR8, 1.4% for SC4 and 2.9% for the joint lesion.

The WR8 lesion at the answer region alone (exploratory O4) cost naming 0.70 [0.34, 1.09] nats. It cost naming with the word shown 0.0008 nats and copying 0.0003 nats. It raised the cue benefit by 1.59 [1.12, 2.11] nats on 132 concepts. The confirmatory recovery values are in Results 3.2.

S3.4. Direct contribution of each member

The direct contribution projects the vector that a component writes at the answer position onto the target's first-token direction, relative to the first tokens of the other concepts' names (Section 2.4). On the 160 test concepts the WR8 sum was 2.28 [1.65, 2.94] logits. Across concepts it correlated .89 with the total first-token effect of the lesion and .55 with the naming loss. The per-member values below come from the 80 calibration concepts and are exploratory. The set sum there was 2.74 [1.74, 3.90] logits, against -0.004 [-0.030, 0.023] for SC4 and 5.71 [4.43, 7.10] for WR30. The per-member values on the 160 test concepts, with descriptive intervals, are in `docs/paper-prep/figdata/fig3b_dla.csv` and are drawn in Fig. S2A. They have no ledger rows.

WR8 member	Direct contribution, logits [95% CI], 80 calibration concepts	Ledger row
expert 43:70	1.03 [0.53, 1.67]	`dla-7dace67f8d88`
expert 45:93	0.81 [0.22, 1.56]	`dla-323d5edbb0ff`
expert 46:61	0.50 [0.24, 0.76]	`dla-7baaa6ec7f39`

WR8 member	Direct contribution, logits [95% CI], 80 calibration concepts	Ledger row
expert 42:115	0.22 [0.14, 0.32]	`dla-a0e6c0bd48e9`
expert 45:32	0.09 [-0.10, 0.30]	`dla-5b4d1833c00b`
expert 46:101	0.08 [-0.15, 0.34]	`dla-d7fffd399685`
head 37:24	0.005 [-0.021, 0.029]	`dla-096a80b38433`
expert 45:48	0.00 [0.00, 0.00]	`dla-920b4b3fcb18`

Expert 45:48 contributed nothing at the answer position on the calibration concepts. The confirmatory run recorded, for each concept, whether the router selected each WR8 expert at the last prompt position and at any position. These counts are in the run's router table, whose SHA-256 is listed in `reports/confirm/optogenetics/result.json`, and they have no ledger rows. The members do not share the set's significance, and the set sum is the tested quantity.

S3.5. Learned searches

A trainable gate α was placed on each of the 2,048 experts and 512 heads of layers 16 to 31, and the gates were optimized with the model's weights fixed. The comprehension search removed the WR8 and WR30 members from its pool. WR8 has no member in these layers, and the five WR30 members removed were head 19:5, expert 21:0, expert 27:61, head 30:14 and head 31:5, leaving 2,555 gates. Each search ran seeds 0, 1 and 2 for 200 steps with sparsity coefficient .0003. A first run with coefficient .03 shrank every gate toward intact and is kept as a calibration record.

The hub objective rewarded three things: larger loss for rare concepts, shared loss across naming, definition choice and synonym choice measured as the per-concept minimum of the three losses, and correlation among the task losses. Units came from the intact model's spread on the screening families. The comprehension objective applied the shared-loss and frequency terms to the two comprehension tasks only and constrained naming loss. The relative objective replaced the naming score by log[exp ℓ(target) + exp ℓ(synonym)] - ℓ(coordinate) and kept everything else. Each seed yielded a soft state, the learned gates themselves, and hard sets of the 4, 8, 16, 32 and 64 most closed components, which are mean-ablated in full. A candidate counted as general damage when neutral-text loss rose by more than 10% or copying or single-word format accuracy fell by more than five percentage points. Candidates showing every predicted direction on the development families ranked first, then the total objective score.

The development selection, saved in `docs/analysis-plan/selections/title-gap-fine.json` and frozen under tag analysis-frozen-title-gap-fine-20261020, chose

hub_s2_learned_soft, hub_s2_learned_hard_4, comprehension_s2_learned_soft and comprehension_s0_learned_hard_64. Only the hub hard set showed the complete predicted direction in development. The hub hard set is experts 24:78, 25:68, 26:102 and 28:115, and the comprehension hard set holds 23 experts and 41 heads listed in `experiments/title_gap/configs/fine-final-core.json`. The relative objective, frozen separately under analysis-frozen-title-gap-relative-hub-20261020, chose relative_hub_s0_learned_soft and relative_hub_s1_learned_hard_64. Each hard set entered the final test with its minus-one states and the next two plus-one states of its ranking. The equal-size comparison used the 64-member hard sets of all three seeds under the hub and the relative objective.

On the final families the neutral-text loss rose by 3.1% under hub_s2_learned_soft and by 1.4% under comprehension_s2_learned_soft. It fell by 0.03% under the hub hard set and by 0.75% under the comprehension hard set. Copying stayed at 1.00 in every state, and format accuracy moved by at most 3.3 percentage points. Generated naming accuracy on the 120 final families was .900 [.842, .950] intact and .767 [.692, .842] under hub_s2_learned_soft. It was .900 under the hub hard set and .892 under both comprehension states. Under relative_hub_s0_learned_soft it was .808 and under relative_hub_s1_learned_hard_64 .800. The 200 matched random sets planned for each chosen state were not run.

A separate learned search, run after the main analyses, aimed at the semantic control effect instead: it selected 32 components, 18 heads and 14 experts, none shared with SC4 (SC32 in Table S2), whose lesion produced a control effect of 14.2 logits on its own held-out items. It is exploratory and is cited in Sections 3.1 and 4.2 only as evidence that control is distributed over more than one set.

S3.6. Broad graded lesions

The broad lesions retained a fixed proportion of the output of all 2,048 experts in a band of sixteen layers. The middle band, layers 16 to 31, was run at .9, .75, .5, .25 and .1. The early band, layers 0 to 15, was run at .5 and the late band, layers 32 to 47, at .75, both carried over from the clinical development trajectories. Full closure of the middle band crossed the format eligibility line in development and was kept as a boundary comparison only. Generated naming accuracy on the final families was .792 [.717, .858] at .1 retained, .850 at .25 and .842 at .5. It was .875 at .75 and at .9. The early band gave .842 and the late band .883. In the progressive cueing test the middle band at .25 and .5 retained named .86 and .90 of the 132 concepts without a cue, and the final cue rescued every failure in both.

S3.7. Weight-masking lesions

A weight-masking lesion replaces a weight matrix W by W $\odot$ M, where each entry of M is drawn from Bernoulli(1 - s) and s is the fraction removed. Surviving

weights are not rescaled. The random stream of each matrix derives from the mask seed and the matrix name, so masks nest across severities within a seed. Router weights and biases are never masked. The first development stage masked layers 4, 12, 20, 28, 36 and 44. It masked two families, the attention family (query, key, value and output matrices together) and the feedforward family (gate, up and down projections of all experts together). Severities .25, .5 and .75 used seeds 1910 and 1911, and severity 1 used the single all-zero mask, 84 states in all. The second stage masked single matrix categories, all seven in layers 4 and 28 and gate, up and down in layer 44. Severity .5 used both seeds and severity 1 one mask, 51 states plus intact and six component references. Nine states were selected on development for the final families. Control items in the second stage were scored in one layout of batch 32, whereas the first stage used the three-layout mean.

Table S6. The nine weight-masking lesions on the 120 final families, each named by layer, matrix, severity and mask seed. Loss is intact minus lesioned, naming in nats with both spellings of each name scored (Section S3.12) and definition choice in logits. Slope is the naming loss per log10 unit of BNC frequency, positive when common words lose more. Differences are mask minus SC4 on the same families. Named is generated naming accuracy. LM is the relative change in neutral-text loss.

State	Naming loss	Definition loss	Slope	Naming minus SC4	Definition minus SC4	Named	LM
L4 query .5 seed 1910	-0.06 [-0.17, 0.05]	0.36 [0.15, 0.58]	0.18 [-0.15, 0.51]	-0.21 [-0.40, -0.03]	-0.58 [-0.83, -0.32]	.900	+0.1%
L4 query .5 seed 1911	-0.01 [-0.10, 0.08]	-0.01 [-0.20, 0.19]	-0.13 [-0.34, 0.06]	-0.15 [-0.30, -0.02]	-0.94 [-1.18, -0.69]	.883	-0.0%
L4 query 1	-0.15 [-0.32, 0.01]	0.80 [0.42, 1.21]	0.20 [-0.19, 0.59]	-0.29 [-0.53, -0.07]	-0.13 [-0.52, 0.29]	.883	+1%
L4 value 1	-0.15 [-0.28, -0.04]	0.17 [-0.11, 0.46]	0.26 [-0.05, 0.62]	-0.30 [-0.48, -0.13]	-0.76 [-1.07, -0.45]	.900	+0.0%
L28 expert gate 1	-0.06 [-0.19, 0.08]	0.41 [0.26, 0.56]	0.24 [-0.09, 0.60]	-0.20 [-0.39, -0.02]	-0.52 [-0.73, -0.31]	.867	+0.0%
L28 expert up .5 seed 1910	0.01 [-0.07, 0.10]	0.33 [0.23, 0.43]	0.12 [-0.07, 0.31]	-0.13 [-0.29, 0.02]	-0.60 [-0.78, -0.42]	.867	+0.0%

State	Naming loss	Definition loss	Slope	Naming minus SC4	Definition minus SC4	Named	LM
L28 expert up .5 seed 1911	0.00 [-0.09, 0.10]	0.42 [0.33, 0.52]	0.25 [-0.03, 0.54]	-0.14 [-0.31, 0.03]	-0.51 [-0.67, -0.34]	.867	+0.0%
L44 expert gate 1	0.18 [-0.34, 0.64]	0.50 [0.33, 0.68]	-0.59 [-1.72, 0.35]	0.03 [-0.48, 0.50]	-0.43 [-0.64, -0.22]	.817	+6%
L28 query 1	0.00 [-0.10, 0.10]	2.46 [2.30, 2.62]	0.00 [-0.27, 0.26]	-0.15 [-0.33, 0.02]	1.53 [1.36, 1.70]	.892	+1%

On the same families WR8 cost naming 0.55 [0.26, 0.89] nats with a slope of -0.63 [-1.59, 0.13], and SC4 cost naming 0.14 [0.01, 0.30] nats and definition choice 0.93 [0.76, 1.09] logits. None of the nine masks lowered naming reliably, so these lesions could not show whether a lesion that impairs naming as much as WR8 does would also produce the frequency effect. The layer-28 query mask was comprehension-selective: it lowered the definition choice by 2.46 logits and the synonym choice by 1.37 [1.22, 1.52] logits while its naming loss was 0.00 [-0.10, 0.10] nats. In the attention formula, zeroing a layer's query or key matrix equalizes the match scores of all context positions. Zeroing its value or output matrix cuts that layer's attention output instead, so the two operations ask different questions. No random comparison was run for the masks.

S3.8. Production control components

PC8 was searched on the homograph development families. Screening used unhinted production under both levels of contextual support and both meanings. The screening score is the log-probability advantage of the context-supported answer over the other meaning's answer, averaged over the two meanings, so a preference for one word cancels. Attribution ranked all components, and the 32 highest were verified by single ablation. A candidate was rejected when it raised neutral-text loss by more than 10% or lowered single-word format accuracy by more than five percentage points. Candidates of 2, 4 and 8 members, whole-sequence and answer-region, at $\alpha = 0$ and .5 were compared on the 20 validation families. PC8 at $\alpha = 0$ over the whole sequence was chosen, with a copying drop of 0.0 on the development families. The final homograph run held 15 core states and 200 matched random sets for PC8 with seed 1910. Under weak context, lesioning PC8 cut the benefit of a correct cue to the word for the intended meaning by 7.15 nats, yet raised that

benefit by 1.45 logits when the model chose between two labeled words, and none of the 200 matched random sets changed either benefit as much. With no cue and weak context, the intact model gave no answer for the other meaning in any of the 40 final families, the PC8 lesion gave none either, and the WR8 lesion gave one in .05 of them. In these conditions .08 to .18 of the 40 answers could not be classified by rule or judge and enter the results as classification bounds. These opposite effects leave open what PC8 computes.

S3.9. Joint states on the final families

The final family run held 100 core states, among them WR8 with SC4 at $\alpha = 0$ and at $\alpha = .5$, and WR8 with the comprehension hard set. Candidate scores used batch sizes 64, 47 and 31, labels 16, 11 and 7, and greedy generation batch 16 with at most eight tokens. The joint lesion with the comprehension hard set named .883 [.825, .942] of the final concepts.

S3.10. Graded lesions of each set (H2f)

H2f asked whether a partial lesion of each set keeps its own marker and leaves the other's unchanged: at 25%, 50% and 75% of WR8 retained, the naming loss should stay above zero and the six-option loss below its margin, and at the same strengths of SC4 the six-option loss should stay above zero and the naming loss below its margin. Table S5 lists the twelve conditions, which were judged together. The test was not supported because the naming loss at 75% of WR8 retained did not exclude zero.

Table S5. The twelve conditions of H2f on the 160 concepts of the naming battery. Losses are intact minus lesioned, with 95% percentile intervals from 10,000 paired bootstrap resamples, and p values are one-sided and Holm-corrected within the third family.

Set	Retained	Condition	Estimate [95% CI]	Holm p
WR8	25%	naming loss above 0	0.40 [0.14, 0.69] nat	.028
WR8	25%	six-option loss below the margin	-0.74 [-0.79, -0.70] logit	.028
WR8	50%	naming loss above 0	0.20 [0.02, 0.39] nat	.028
WR8	50%	six-option loss below the margin	-0.57 [-0.61, -0.54] logit	.028
WR8	75%	naming loss above 0	0.08 [-0.00, 0.17] nat	.028
WR8	75%	six-option loss below the margin	-0.33 [-0.35, -0.31] logit	.028
SC4	25%	six-option loss above 0	1.01 [0.91, 1.10] logit	.028
SC4	25%	naming loss below the margin	0.02 [-0.02, 0.06] nat	.028

Set	Retained	Condition	Estimate [95% CI]	Holm p
SC4	50%	six-option loss above 0	0.50 [0.43, 0.57] logit	.028
SC4	50%	naming loss below the margin	0.01 [-0.02, 0.04] nat	.028
SC4	75%	six-option loss above 0	0.22 [0.18, 0.26] logit	.028
SC4	75%	naming loss below the margin	0.01 [-0.01, 0.02] nat	.028

S3.11. Group-level comparison of two accounts (Section 3.3)

To ask whether the two disorders differ in kind or only in severity, we fitted a severity-only account, in which the two patient groups differ only in the strength of a single lesion, and a two-lesion account, in which the strengths of the WR8 and SC4 lesions vary separately. Each account was fitted to two of the three marker groups and judged by its mean squared error on the third. The comparison used the three marker groups of the patient reanalysis, the frequency effect, progressive cueing and the naming error types (Table S7), and the fixed lesion states of Section S3.3. In the severity-only account one lesion varies in strength and the two patient groups receive different strengths. In the two-lesion account the WR8 and SC4 strengths vary separately. For each held-out marker group, the strengths that best matched the other two groups on development material were fixed and then predicted the held-out group for SD and SA. Each marker was placed on the calibrated scale of Section 2.4: the frequency effect, for example, is the high-minus-low accuracy gap divided by the intact model's distance from chance. The error of an account is the mean squared difference between predicted and observed values, with equal weight on the two patient groups and the marker groups. Overall errors were 0.104 [0.083, 0.157] for two lesions and 0.098 [0.079, 0.143] for severity only, a difference of 0.006 [-0.035, 0.047]. By marker group (two lesions / severity only) the errors were 0.239 / 0.233 for cueing, 0.053 / 0.042 for the frequency effect and 0.019 / 0.019 for the error types. The cueing group carries most of the error because every lesion state was rescued more often than patients with SD. The calibration was fixed in advance and stands in for a common scale that has not been validated for the model and the patients.

The strengths were chosen on development material from measured states only, listed in `experiments/title_gap/configs/clinical-curves-extended-dev.csv`. The severity-only account had one trajectory per kind of lesion: WR8 at $\alpha = 1$, .5 and 0, SC4 at the same three values, the middle band of layers 16 to 31 at $\alpha = 1$, .9, .75, .5, .25 and .1, the early band at 1 and .5, the late band at 1 and .75, and eight weight-masking trajectories holding the nine masks of Table S6 with the intact state, at severity .5 and 1 where both were measured. The two-lesion account used the measured grid of WR8 and SC4 at $\alpha = 1$, .5 and 0 each, nine states.

For each held-out marker group, each patient group took the state on a trajectory that minimized its squared error over the two training groups, with equal weight per group, and the trajectory with the lowest mean of the two patient groups' training errors was kept, ties going to the first trajectory and state in name order. Nothing was interpolated between measured states, and the 256 figure-precision scenarios each carried their own development selection. The development selection, frozen in `docs/analysis-plan/selections/title-gap-clinical.json` under tag analysis-frozen-title-gap-clinical-v2-20261019, was transferred unchanged to the final items. With cueing held out, two lesions placed SD at the intact state and SA at WR8 fully ablated with SC4 at $\alpha = .5$, and severity only placed SD at the middle band at $\alpha = .5$ and SA at the middle band at .25. With the frequency effect held out, two lesions placed SD at WR8 fully ablated and SA at WR8 and SC4 both fully ablated, and severity only placed both groups at the middle band at .1. With the error types held out, both accounts placed both groups at WR8 fully ablated, on the WR8 trajectory under severity only. The final run therefore measured eight states: intact, SC4, WR8, WR8 with SC4 at $\alpha = 0$ and at .5, and the middle band at .1, .25 and .5. Development used 36 calibration concepts for cueing and error types and the 43 development families for the frequency marker, and the final test used the 132 test concepts and the 120 final families. The development errors were .113 for two lesions and .147 for severity only, exploratory.

S3.12. Scoring the family names in both spellings

The naming prompts of the final families end with "Reply with one word only" and, unlike those of the naming battery, neither ask for a lowercase answer nor open with demonstration items. The frozen family protocol scored the lowercase target only. Scored in both spellings, the intact model gave .70 of the name's probability to the capitalized spelling, the SC4 lesion .76, the WR8 lesion .75 and the learned hub lesion .995. A lesion that moves probability toward the capitalized spelling therefore lowers the lowercase score without any change in which word the model names. We scored every candidate again with its first letter capitalized, in the execution snapshots of the final runs and with the same states, batch layouts and means (`experiments/title_gap/rescore_case.py`), and used log[P(word) + P(Word)] as the naming score of every family analysis (`experiments/title_gap/analyze_case.py`). The rescoring reproduced the stored lowercase scores of the intact, WR8 and SC4 states exactly. `reports/explore/2026-11-02-case-rescore/states.csv` lists both scores for all 110 states.

A separate check crossed the prompt format with the item set, scoring the 160 battery concepts without the demonstration items and the 120 families with them, for the intact model, WR8 and SC4 (`experiments/e3_errors/run_demo_`

format.py). SC4 lowered naming by 0.03 [-0.01, 0.08] nats on the battery with the demonstration items and by -0.02 [-0.08, 0.04] without them. On the families it lowered the lowercase score by 0.05 [-0.09, 0.19] with the demonstration items, which made the model answer in lowercase (.012 of the probability on the capitalized spelling), and by 0.79 [0.62, 0.96] without them, or 0.15 [0.01, 0.30] with both spellings scored. WR8 lowered naming in every cell: 0.84 [0.45, 1.27] and 0.61 [0.24, 1.00] on the battery with and without the demonstration items, 0.44 [0.13, 0.76] on the families with them and 0.55 [0.26, 0.90] without them, both spellings scored. The SC4 naming cost on the families therefore came from the case of the answer, not from the concepts or the demonstration items. Under the learned hub lesion the naming loss was 1.56 [0.99, 2.15] nats and the partial correlation of naming and definition-choice losses .24 [.07, .40].

S4. Patient data

Patient values were transcribed on 2026-10-13 from the published papers into data/literature/. Each row names its source table, figure or results section, and whether it was copied from a table, copied from the text or read from a figure. Every row was checked against the originals the same day, and no transcription error was found. The individual patient table jefferies2006_individual.csv was transcribed from Tables 4 and 5 of the original PDF on 2026-10-19, with the DOI, file hash and conversion rules in the adjacent JSON. Three values printed as 100 in Table 5 were read as 1.00 within a table of proportions. Two patients of the stroke group (KA and MS) are absent from Table 5. The individual cueing values of Jefferies et al. (2008) were measured on 2026-10-28 from the bars of their Fig. 2, which give each patient's correct responses out of 30 at each cue. Every value lies within 0.0025 of a multiple of 1/30, and the patients' values reproduce the published group averages (data/literature/jefferies2008_cueing_individual.csv).

Table S7. Patient values used in the paper and their sources. Reading precision is the transcription precision of a figure estimate, not patient sampling error. The synonym judgment difference is the mean over imageability levels. The calibrated gap is the accuracy difference divided by the healthy to chance distance, with healthy accuracy .95 and three choices, as fixed in data/literature/clinical_calibration.json. Rescue is (cued minus uncued)/(1 minus uncued), and the final cue is the whole word minus its last phoneme. Error shares are proportions of single-word semantic errors. Rows names the file in data/literature/, H11 for hoffman2011_frequency.csv, J06 for jefferies2006_markers.csv and J08 for jefferies2008_cueing.csv, or the ledger for derived values.

Marker	Group, n	Value	Source	How read	Precision	Rows
Synonym judgment, high minus low frequency	SD, 13	.233	Hoffman et al. (2011), Fig. 3	figure estimate, difference	about .02 per cell	H11
Same	SA, 13	.000	same	same	same	H11
Frequency effect, calibrated gap	SD, 13	.378 (figure bounds .313 to .443)	same, with the calibration file	derived	bounds listed	ledger
Same	SA, 13	.000 (figure bounds -.064 to .065)	same	derived	bounds listed	ledger
Frequency effect in synonym judgment	SD	$t(12) = 5.28$, $p < .001$	Hoffman et al. (2011), Results	text		H11
Same	SA	$t(12) = 0.18$	same	text		H11
Picture naming, high and low familiarity	SD, 10	45 and 30 % correct	Jefferies & Lambon Ralph (2006), Fig. 2	figure estimate	about 2 points	J06
Same	SA, 10	33 and 31 % correct	same	same	same	J06
Uncued naming, 30 pictures	SD, 8	.338	Jefferies et al. (2008), Fig. 2	figure estimate, mean of patient bars	.002	J08
Same	SA, 6	.456	same	same	same	J08
Naming with a two-phoneme cue	SD, 8	.454	same	same	same	J08
Same	SA, 6	.767	same	same	same	J08
Naming at the final cue	SD, 8	.579	same	same	same	J08
Same	SA, 6	.944	same	same	same	J08

Marker	Group, n	Value	Source	How read	Precision	Rows
Rescue by the two-phoneme cue	SD, 8	.175	derived from the three rows above	derived	bounds .170 to .181	ledger
Same	SA, 6	.572	same	derived	bounds .566 to .577	ledger
Rescue by the final cue	SD, 8	.364	same	derived	bounds .359 to .369	ledger
Same	SA, 6	.897	same	derived	bounds .893 to .901	ledger
Cue benefit larger in SA, two-phoneme cue	SA vs SD, 14	F(1,12) = 8.0, p = .02	Jefferies et al. (2008), Results	text		J08
Coordinate share of semantic errors	SD, 10	.74 [.58, .88]	Jefferies & Lambon Ralph (2006), Table 5	mean of individual proportions		ledger
Superordinate share	SD, 10	.25 [.11, .41]	same	same		ledger
Associative share	SD, 10	.01 [0, .04]	same	same		ledger
Coordinate share	SA, 8	.54 [.44, .65]	same	same		ledger
Superordinate share	SA, 8	.18 [.06, .33]	same	same		ledger
Associative share	SA, 8	.27 [.16, .39]	same	same		ledger
Associative share, SA minus SD	18	.26 [.14, .38]	same	same		ledger
Associative errors, group test	SA vs SD	t(16) = 4.58, p = .0003	Jefferies & Lambon Ralph (2006), Table 5 note	text		J06

Marker	Group, n	Value	Source	How read	Precision	Rows
Ease of rejecting distractors against accuracy	SD, 10	r = .24, p = .06	Jefferies & Lambon Ralph (2006), Results	text		J06
Same	SA, 10	r = .50, p < .001, factor by group Wald = 9.20	same	text		J06
Familiarity effect within group	SD	every task, p < .025	same	text		J06
Same	SA	no task, t < 1	same	text		J06
Phonemic cue, Boston Naming Test	SD, 3	4.5 to 7 % correct	Jefferies & Lambon Ralph (2006), Fig. 4	figure estimate	about 2 points	J06
Same	SA, 8	25.5 to 52 % correct	same	same	same	J06

Three points qualify these rows. The cued naming of Jefferies and Lambon Ralph (2006), Fig. 4, comes from three other patients with SD (EK, BS and GE), not the ten of Tables 4 and 5. The cueing values of Jefferies et al. (2008) were read from the author manuscript (White Rose 3725), whose figure bars fall on multiples of 1/30 and whose averages match the printed group bars within .002. The published version could not be downloaded (HTTP 403), so the manuscript values stand. Its six patients with SA are six of the ten of the 2006 study. The group-level comparison of Section 2.4 used the frequency, cueing and error families above. Patient error cases were resampled, and the frequency and cueing group means stayed fixed, so those intervals are conditional on the published means. The cumulative rescue of the model follows the same formula, with the first correct response at any cue length counted and the uncued failures of the same state as denominator. The file `tms_virtual_lesion.csv` (Pobric et al., 2007; Whitney et al., 2011) was transcribed for the earlier scorecard and is not used in the paper.

S5. Hypothesis numbering and freeze chronology

Every decision rule was fixed and tagged in Git before the lesion results on the corresponding held-out items were examined. No public preregistration was made. Three qualifications are recorded. The control items had been inspected intact during model selection. The battery's test concepts had intact validity scores before their lesion results were opened. At the graded and positional freeze, the intact, WR8, SC4 and SC4 answer-region results on those concepts were already known, and only the new strengths, amplifications, other regions and joint lesions were new conditions. The answer-region part of H4e repeats the earlier H4d comparison.

Table S3. The fourteen staged freezes. Dates are the tag dates in local time. Tags named analysis-frozen-session... and the v1 versions of the October 19 tags record engineering checkpoints or superseded protocol versions and are not freezes of decision rules. Plan files are under `docs/`.

Freeze	Git tag	Date	What it sealed	Plan file
Original inclusion criterion	g1-frozen	2026-09-27	Rules, materials and analysis before any real model ran on the control items	`plans/pilot-protocol.md`
Backup criterion	g1-backup-frozen	2026-09-27	Backup model OLMoE-0125 registered, rules unchanged	`plans/pilot-protocol.md`
Revised inclusion criterion	g1-revised-frozen	2026-09-28	Order-averaged scoring, before the Qwen component search	`adr/0003-qwen-and-revised-g1.md`
Control confirmation	analysis-frozen-control	2026-09-28	H1a rules, before the control test items were opened	`analysis-plan/control-test.md`
Retrieval confirmation	analysis-frozen-knowledge	2026-09-29	H1b rules and WR8, before the naming test split was opened	`analysis-plan/knowledge-test.md`

Freeze	Git tag	Date	What it sealed	Plan file
Battery primary and secondary tests	analysis-frozen-experiment3	2026-10-08	H3a, H3b, S1 to S6, S7 descriptive, before any lesion on the test concepts	`analysis-plan/experiment3.md`
Graded and positional tests	analysis-frozen-optogenetics	2026-10-13	R1 to R6, 45 states, commit 750d8a4	`analysis-plan/optogenetics.md`
gpt-oss revised format	analysis-frozen-p12b	2026-10-16	Revised format check for the second-model census	`plans/second-model-g1-p12.md`
P14 reading comparisons	analysis-frozen-p14-reading	2026-10-17	Reading rules of the learned-gate exploration (high_demand required for color and size only)	`plans/p14-learned-gates.md`
Coder replication gate	analysis-frozen-title-gap-coder-v2-20261019	2026-10-19	Census and replication rules for Qwen3-Coder, revised freeze	`analysis-plan/title-gap-coder.md`
Homograph confirmation	analysis-frozen-title-gap-homonyms-v2-20261019	2026-10-19	15 core states, PC8 random sets, classification bounds, file hashes bound	`analysis-plan/title-gap-homonyms.md`
Clinical transfer	analysis-frozen-title-gap-clinical-v2-20261019	2026-10-19	Development-selected states and the two-account comparison, patient source bytes preserved	`analysis-plan/title-gap-clinical.md`
Semantic family confirmation	analysis-frozen-title-gap-fine-20261020	2026-10-20	100 core states, learned and broad and masked lesions, comparisons and resampling	`analysis-plan/title-gap-fine.md`

Freeze	Git tag	Date	What it sealed	Plan file
Relative naming objective	analysis-frozen-title-gap-relative-hub-20261020	2026-10-20	Relative-objective candidates and the equal-size comparison	`analysis-plan/title-gap-relative-hub.md`

The paper numbers its predictions H1 to H4 by the four questions of the Introduction (Table S8). The historical labels under which each test was frozen are kept in the analysis plans and in the ledger, and the families within which Holm correction was applied did not change with the renumbering. The primary family holds the original H3a and H3b. The secondary family holds S1 to S6, and S7 was excluded under its frozen development rule and is descriptive. The third family holds R1 to R6. The inclusion criterion is not counted among the predictions.

Table S8. Paper labels, historical labels and verdicts.

Paper label	Content	Historical label	Family	Verdict
inclusion criterion	Intact control effect and high-demand accuracy	original H1, G1	none	passed for the main model
H1a	Selective control damage, with the similarity slope as a falsification condition	control main effect and association slope	control confirmation	supported
H1b	Double dissociation of frequency and control effects	original H2, ΔF and ΔC	retrieval confirmation, intersection-union	supported
H2a	Cross-format dissociation of naming and selection	S1	secondary	supported
H2b	Naming impaired, comprehension unchanged	S2	secondary	supported

Paper label	Content	Historical label	Family	Verdict
H2c	Direct contribution to the target's first token	S4	secondary	supported
H2d	Lesion strength changes each set's own marker	R1	third	supported
H2e	Answer-region recovery of naming and cue benefit	R4	third	supported
H2f	Markers stay specific at every strength	R2	third	not supported
H2g	Joint lesion interaction	R6	third	supported
H3a	Error types beyond general weakening	H3a	primary	supported, below the smallest effect size of interest
H3b	Larger retrieval set keeps errors within the category	S5	secondary	not supported
H4a	Cue dependence, SC4 greater than WR8	H3b	primary	not supported, direction reversed
H4b	Retrieval loss predicts cue dependence	S6	secondary	supported
H4c	Amplifying WR8 lowers cue dependence	R3	third	supported
H4d	Answer-region control lesion raises miscue cost	S3	secondary	supported
H4e	Middle and answer regions reverse the miscue effect	R5	third	supported

Paper label	Content	Historical label	Family	Verdict
none	Contextual intruder words	S7	descriptive	not entered
none	Alternative sets, learned searches, P14, scorecard	exploratory reports	none	exploratory

S6. Component members

Table S2. Members of the component sets, with the repository codes used in the analysis code and outputs. Indices are zero based, written layer:index. Sources: `docs/paper-prep/component-members.csv`, `docs/analysis-plan/selections/title-gap-production.json` (PC8), `reports/explore/qwen_new_sets.json` (SC4b), and the run records copied to `docs/analysis-plan/selections/k8b-selection.json` (WR8b) and `docs/analysis-plan/selections/p14-control-groups.json` (SC32).

Paper code	Repository code	Members	Selection	Overlaps
WR8	K8	experts 46:61, 46:101, 45:32, 45:48, 43:70, 42:115, 45:93 and head 37:24	Naming screening set, verified shortlist, expanded from WR4 (experts 46:61, 46:101, 45:32 and head 37:24)	none with SC4
SC4	C4	heads 9:15, 27:9, 25:24, 24:5	Control screening items after the domain-general exclusion	none with WR8
WR8b	K8b	heads 33:16, 40:14, 40:15, 31:5, 38:5, 38:3 and experts 41:62, 45:119	WR8's rule with language-loss and scaling exclusions	none with WR8, all eight lie within WR30, expert 41:62 shared with PC8

Paper code	Repository code	Members	Selection	Overlaps
SC4b	C4s	head 9:15, head 14:13, expert 35:28, head 6:1	Alternative control set	head 9:15 shared with SC4, the other three lie within SC16
PC8	PC8	heads 47:21, 43:29, 47:28, 41:21 and experts 47:2, 47:101, 47:125, 41:62	Homograph development families, Section S3.8	expert 41:62 shared with WR8b and WR30
SC32	P14 control	heads 46:23, 45:25, 44:1, 44:28, 27:31, 32:16, 35:19, 27:30, 24:19, 9:10, 21:21, 12:11, 35:15, 3:7, 35:2, 39:18, 19:1, 35:10 and experts 34:57, 17:88, 30:8, 46:71, 39:93, 43:96, 3:41, 38:67, 6:53, 18:71, 25:103, 25:104, 2:82, 6:35	Learned-gate exploration P14, control objective, k = 32	none with SC4, not a prefix of the SC ranking

The larger prefixes of the two rankings are listed here. WR16 adds to WR8 the heads 46:3, 41:12, 33:16, 38:5, 40:15, 14:13 and the experts 45:12, 41:79. WR30 adds to WR16 the heads 38:3, 31:5, 19:5, 40:14, 32:4, 30:14, 6:14 and the experts 45:116, 41:62, 21:0, 27:61, 45:119, 44:22, 2:85. SC8 adds to SC4 the heads 14:13, 33:20, 21:19 and the expert 35:28. SC16 adds to SC8 the heads 6:1, 26:16, 32:6, 21:3, 9:3, 22:13, 26:13, 22:17. The hub hard set of the learned search is experts 24:78, 25:68, 26:102 and 28:115. The P14 knowledge and hub candidates (24 and 8 components) keep their repository names.

References

Gilbert, R. A. and Rodd, J. M. (2022). Dominance norms and data for spoken ambiguous words in British English. *Journal of Cognition* 5, 4. doi:10.5334/joc.194

Hoffman, P., Rogers, T. T., and Lambon Ralph, M. A. (2011). Semantic diversity accounts for the "missing" word frequency effect in stroke aphasia: Insights using a novel method to quantify contextual variability in meaning. *Journal of Cognitive Neuroscience* 23, 2432–2446. doi:10.1162/jocn.2011.21614

Jefferies, E. and Lambon Ralph, M. A. (2006). Semantic impairment in stroke aphasia versus semantic dementia: A case-series comparison. *Brain* 129, 2132–2147. doi:10.1093/brain/awl153

Jefferies, E., Patterson, K., and Lambon Ralph, M. A. (2008). Deficits of knowledge versus executive control in semantic cognition: Insights from cued naming. *Neuropsychologia* 46, 649–658. doi:10.1016/j.neuropsychologia.2007.09.007

Pobric, G., Jefferies, E., and Lambon Ralph, M. A. (2007). Anterior temporal lobes mediate semantic representation: Mimicking semantic dementia by using rTMS in normal participants. *Proceedings of the National Academy of Sciences* 104, 20137–20141. doi:10.1073/pnas.0707383104

Whitney, C., Kirk, M., O'Sullivan, J., Lambon Ralph, M. A., and Jefferies, E. (2011). The neural organization of semantic control: TMS evidence for a distributed network in left inferior frontal and posterior middle temporal gyrus. *Cerebral Cortex* 21, 1066–1075. doi:10.1093/cercor/bhq180

Wu, W. and Hoffman, P. (2022). Validated measures of semantic knowledge and semantic control: Normative data from young and older adults for more than 300 semantic judgements. *Royal Society Open Science* 9, 211056. doi:10.1098/rsos.211056

Supplementary figures

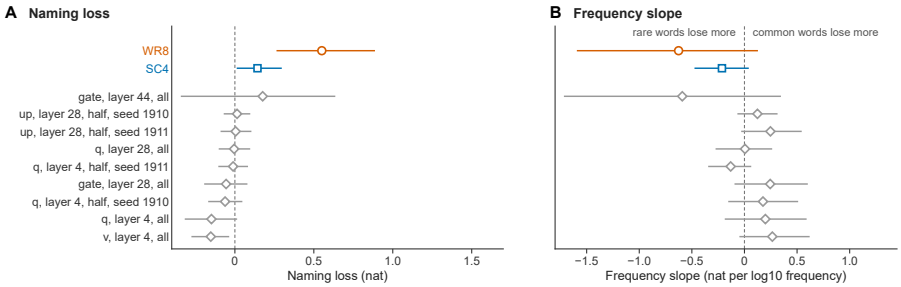


Fig. S1. Weight-masking lesions. (A) Naming loss, with both spellings of each name scored (Section S3.12), and (B) its slope per log10 unit of corpus frequency for WR8, SC4 and the nine weight-masking lesions on the 120 final families. Each mask zeroes all entries or a random half of one weight matrix in one layer and is named by matrix, layer, share and seed. A positive slope means that common words lose more. Error bars are 95% bootstrap intervals over families.

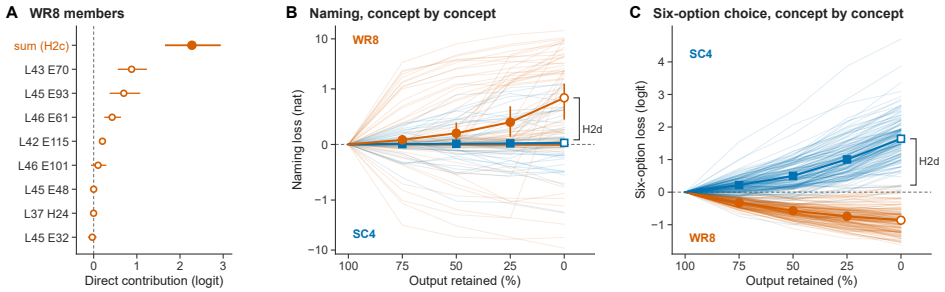


Fig. S2. WR8 members and graded lesions, concept by concept. (A) Direct contribution of each WR8 member, named by layer and expert (E) or attention head (H), and of the whole set (H2c) to the first token of the name. (B) Naming loss and (C) six-option loss as WR8 or SC4 output is reduced to 75, 50, 25 and 0%, one line per concept, with the means (H2d). All panels use 160 concepts. Error bars are 95% bootstrap intervals over concepts, descriptive for the members. The naming axis in (B) is logarithmic beyond ±1 nat.

Table S4. Prespecified tests and their outcomes.

Test and condition	Estimate [95% CI]	Unit	n	Holm p	Verdict
H1a Control demand effect of SC4					
high minus low demand loss	4.05 [2.41, 5.66]	logit	63	interval rule	supported
slope on standardized distractor similarity	2.13 [1.28, 2.91]	logit	63	interval rule	supported
H1b Double dissociation					
frequency effect, WR8 minus SC4	1.28 [0.65, 1.94]	nat	240	interval rule	supported
control effect, SC4 minus WR8	4.14 [2.43, 5.80]	logit	63	interval rule	supported
H2a Cross-format dissociation					
naming loss, WR8 minus SC4	0.81 [0.41, 1.24]	nat	160	<.001	supported
six-option loss, SC4 minus WR8	2.50 [2.39, 2.62]	logit	160	<.001	supported
H2b Naming impaired, comprehension unchanged					
naming loss	0.84 [0.45, 1.26]	nat	160	<.001	supported
word-to-definition loss (noninferiority margin 0.73)	−0.01 [−0.10, 0.09]	logit	160	<.001	supported
H2c Direct contribution of WR8					
direct contribution to the first token	2.28 [1.65, 2.94]	logit	160	<.001	supported
H2d Graded lesion strength					
WR8 naming, complete minus 75% retained	0.76 [0.42, 1.13]	nat	160	<.001	supported
SC4 six-option, complete minus 75% retained	1.42 [1.34, 1.50]	logit	160	<.001	supported
H2e Recovery at the answer positions					
naming, complete minus 75% restored	0.62 [0.31, 0.95]	nat	160	<.001	supported
cue benefit, complete minus 75% restored	1.45 [0.96, 2.02]	nat (restricted log odds)	100	<.001	supported
H3a Error types beyond general weakening					
shift toward associates, SC4 minus WR8	.06 [.03, .09]	five-candidate probability	156	.012	supported, below the smallest effect size of interest
excess over the weakening model	.04 [.01, .08]	five-candidate probability	156	.012	supported, below the smallest effect size of interest
H3b Associative errors of the larger retrieval set					
excess over the weakening model	−.01 [−.07, .03]	five-candidate probability	160	.270	not supported
H4a Cue dependence, SC4 versus WR8					
cue dependence, SC4 minus WR8	−2.24 [−3.28, −1.33]	nat (restricted log odds)	100	1.000	not supported
excess over the weakening model	−19.06 [−21.97, −15.61]	nat (restricted log odds)	100	1.000	not supported
H4b Cue dependence versus naming loss					
correlation across 100 concepts	.54 [.37, .68]	Spearman r	100	<.001	supported
H4c Cue dependence after amplifying WR8					
change at twice the WR8 output	−3.41 [−4.15, −2.70]	nat (restricted log odds)	100	<.001	supported
H4d Miscue effect of SC4 at the answer positions					
cue benefit minus miscue cost	−0.44 [−0.60, −0.29]	nat (restricted log odds)	100	<.001	supported
miscue cost	0.40 [0.31, 0.49]	nat (restricted log odds)	100	<.001	supported
H4e Miscue effect by lesion position					
miscue cost, middle region	−0.59 [−0.75, −0.43]	nat (restricted log odds)	100	<.001	supported
miscue cost	0.43 [0.33, 0.53]	nat (restricted log odds)	100	<.001	supported
H2f Each strength keeps its own marker					
twelve conditions (Table S5)				interval rule	not supported
H2g Joint lesion interaction					
six-option interaction	0.37 [0.32, 0.42]	logit	160	<.001	supported
naming interaction (equivalence within ±0.20)	−0.03 [−0.07, 0.01]	nat	160	<.001	supported

Note. Each bold row names a test and its prediction, and the rows under it are its conditions. A conjunctive test (H1b, H2a, H2e, H4d, H2g) requires every condition. Estimates are given with 95% percentile intervals from 10,000 paired bootstrap resamples, and n is the number of items or concepts resampled. p values are one-sided and Holm-corrected within their test family. Interval rule: the test was decided by a prespecified interval decision fixed in advance, not by a p value (Section 2.4). H2f lists its twelve conditions in Table S5. Restricted log odds: the log odds of the target against the associate (Section 2.4). Nats are units of natural log probability and logits are differences of the model's raw scores (Section 2.4). WR8, the eight word retrieval components (seven experts and one attention head). SC4, the four semantic control components (attention heads).